

The project involves creating an AI-based vision program for UAV identification. It may be used in military or conflict environments, so the system must be robust, secure, and capable of operating in difficult conditions.

The main challenges to analyze are:

1. Data privacy and confidentiality. The images and video data are owned by military clients or defense companies. They may not want to send sensitive data to Embention or any external server.
2. Offline operation and edge deployment. The AI system must be integrated into the device with the camera or its onboard computer so it can operate without internet access. The device may face GPS denial, communication jamming, bandwidth limitations, latency constraints, and unstable links. We need to design an architecture that works offline, with local inference, optimized models, fail-safe behavior, and degraded-mode operation.
3. Image quality. The images sent by third party companies of specific flying objects to detect will have different quality than the images taken by the device of UAV in the air.
4. Reliability, safety, and operational robustness. The system must handle bad weather, dust, smoke, night conditions, camouflage, occlusion, motion blur, low resolution, adversarial deception, sensor noise, and changing sky conditions. It must include uncertainty estimation, confidence thresholds, fallback modes, logging, explainability, and human review where appropriate.
5. Security and compliance. The images uploaded by third party companies into the device must not travel to Embention, they must stay inside the device.

The objective of this project now is this: The third party company will give our drone small quantity of images (this quantity will be decided by us through tests) of the same UAV object from different angles. The device must memorize this object so if it sees it in the air, it can identify and for example if our device is a drone, it will fly towards the uav and crash. So let's say that the third-party company needs the device to identify an UAV, so it sends some images of it, and if the device sees it in the sky it let's us know. Also, the hardware doesn't matter much now, I was told that the adaptation will happen later. The important part is the pipeline flow of the embedding of the target and the privacy of the third-party company.

There will be an app that is installed in client's system so that the images they upload don't go through Embention.

The images have to be processed locally.

The expected detection distance is 50 to 100m

We have to identify uav planes approximately 2x1m size. The number of images they have to upload has to be decided by us through tests, to see which is the best number of images and what kind of images.

For the Kai that uses Embention, we would start with 1920×1080 resolution. These are the target sensors we currently have available for local testing, and they provide a reasonable balance between the frame-rate limits we can process and target distance. For the lens HFOV, I would consider a range between 30° and 45°. However, for the detection and engagement analysis, we should not limit ourselves to these ranges. We should use whatever material is available to evaluate an appropriate strategy. The HFOV constraint will limit us during field testing, but not during the analysis which for now is the most important task.

We will address the optimization strategy on real hardware later. That decision depends on the trade-off between the target FPS rate and the model size. For example, if the model is small enough, and/or the quantization/compression is adequate, we may not need to degrade the image. We would crop it and process the full image in overlapping patches. In fact, we are not considering processing the entire image directly. As discussed before, the optimal approach is to feed the engagement model with the detections from a generic detection model, for example YOLO. Therefore, we would only pass the object crops produced by the detection model.